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Cambridge IGCSE Combined and Co-ordinated Sciences

B5 Plant Nutrition

Comprehensive Exam Preparation Guide

B5 Plant Nutrition — Cambridge IGCSE Biology Study Guide

Chapter Overview

In this chapter you will:

- Learn about photosynthesis and the equation for it
- Find out how plants use the carbohydrates made in photosynthesis
- Learn how the structure of a leaf is adapted for photosynthesis
- Investigate the need for chlorophyll, light and carbon dioxide in photosynthesis
- Investigate how light and temperature affect the rate of photosynthesis
- Find out how a plant uses glucose produced in photosynthesis
- Learn about the functions of nitrate ions and magnesium ions in a plant
- Identify structures of the leaf in diagrams and images of a dicotyledonous plant
- Learn how the features of a leaf are adaptations for photosynthesis

Important Terms

Term	Definition
Nutrition	Taking in useful substances (nutrients) that an organism uses to make new parts, repair old parts or release energy
Photosynthesis	The process by which plants synthesise carbohydrates from raw materials using energy from light
Organic substances	Carbon-containing biological molecules such as carbohydrates, fats and proteins; plants can make them but animals must take them in as food
Raw materials of photosynthesis	Carbon dioxide (from the air) and water (from the soil), which react to make glucose and oxygen
Chlorophyll	A green pigment in chloroplasts that captures energy from sunlight and transfers it into energy in carbohydrates
Chloroplast	The part of a plant cell where photosynthesis happens; contains chlorophyll (spread out on membranes) and the enzymes for photosynthesis
Enzymes	Biological catalysts that help the reactions of photosynthesis take place; their activity is affected by temperature
Glucose	A sugar ($C_6H_{12}O_6$) made in photosynthesis; it contains some of the energy that was originally in sunlight
Starch	An insoluble carbohydrate made by linking many glucose molecules into long, spiral chains; the plant's energy store
Starch grains	Clumps of millions of starch molecules inside plant cells, e.g. in potato tuber cells
Sucrose	A sugar whose molecules are made of glucose and another similar molecule (fructose) linked together; the form in which plants transport

	carbohydrate
Cellulose	A carbohydrate made of glucose molecules linked in long, straight chains; used to build plant cell walls
Glycogen	The storage carbohydrate of animals, made of chains of glucose molecules (animals do not make or store starch)
Amino acids	The building blocks of proteins; plants make them from glucose plus nitrogen taken in as nitrate ions
Nitrate ions	Mineral ions absorbed from the soil through the root hairs; they supply the nitrogen needed to make amino acids, proteins and chlorophyll
Magnesium ions	Mineral ions absorbed from the soil; magnesium is needed to make chlorophyll
Nectar	A sweet liquid secreted by many insect-pollinated flowers, to attract their pollinators
Pollen grains	Small structures which contain the male gametes of a flower
Pollinator	An animal (insect, bat or bird) that carries pollen grains from one flower to another
Respiration	The process in cells that releases energy from glucose, e.g. for active transport and for building proteins
Active transport	Movement of particles into or out of a cell against a concentration gradient using energy from respiration, e.g. mineral ions into root hairs
Diffusion	The net movement of particles from a region of higher concentration to a region of lower concentration; how carbon dioxide enters a leaf
Osmosis	The net movement of water molecules through a partially permeable membrane from a dilute solution to a more concentrated solution
Xylem vessels	Tubes that carry water and mineral ions from the roots up to the leaves
Phloem tubes	Tubes that carry sucrose (and amino acids) from one part of a plant to another
Vascular bundles	Collections of xylem tubes and phloem vessels running side by side, which form the veins in a leaf
Lamina	The broad, flat part (blade) of a leaf
Midrib	The main vein running down the centre of a leaf
Leaf stalk	The stalk that joins the lamina to the rest of the plant; it contains vascular bundles
Dicotyledonous (dicot) plant	A flowering plant whose leaves usually have a branching network of veins
Transverse section (TS)	A thin slice cut across a structure, such as a leaf, so that its layers of cells can be seen under a microscope
Cuticle	A thin, transparent, waterproof covering made of a waxy substance, on the upper surface of a leaf
Upper epidermis	The upper layer of tightly packed, transparent cells that reduces water vapour loss and lets light through

Palisade mesophyll	A layer of tall, narrow cells with very large numbers of chloroplasts near the top of the leaf; the main site of photosynthesis
Spongy mesophyll	A layer of photosynthesising cells with many air spaces between them, allowing gases to diffuse through the leaf
Air spaces	Gaps between spongy mesophyll cells through which carbon dioxide, oxygen and water vapour diffuse
Lower epidermis	The bottom layer of cells of a leaf; it contains the stomata and usually has no cuticle
Stoma (plural stomata)	An opening in the lower epidermis that lets carbon dioxide and oxygen diffuse in and out, and water vapour diffuse out
Guard cells	A pair of cells (containing chloroplasts) surrounding each stoma; they change shape to open and close it
Variegated leaf	A leaf that has some white areas (no chlorophyll) and some green areas (with chlorophyll)
Destarching	Leaving a plant in the dark for long enough for it to use up its starch stores
Control	A standard sample that you use as a comparison, to find the effect of changing a variable
Iodine solution	An orange-brown solution that turns blue-black when starch is present
Potassium hydroxide solution	A solution that absorbs carbon dioxide; used to remove carbon dioxide from the air around a leaf
Sodium hydrogencarbonate solution	A solution that provides carbon dioxide for aquatic plants; a higher concentration supplies more carbon dioxide
Rate of photosynthesis	How much photosynthesis happens in unit time, e.g. the number of oxygen bubbles released by an aquatic plant in one minute
Limiting factor	The factor in shortest supply (light intensity, carbon dioxide concentration or temperature) that stops the rate of photosynthesis going any higher
Heat shield	A transparent barrier between a lamp and a plant that lets light through but stops the lamp's heat changing the temperature
Glasshouse	A glass building in which temperature, light and carbon dioxide can be controlled so that crops photosynthesise quickly and give a high yield
Biofuels	Fuels made from plants grown for the purpose, which can be burnt to produce electricity or move vehicles

B5.01 Photosynthesis

Before You Start

The chapter opens with three questions to discuss with a partner, about where the energy in food comes from.

Q: (1) What types of nutrient does your body use to obtain energy?

A: Mainly **carbohydrates** (such as glucose and starch) and **fats** (lipids). Proteins can also be used to release energy, but the body mainly uses them for growth and repair.

Q: (2) Energy cannot be created or destroyed. Where did the energy in the nutrients come from originally?

A: From the **Sun**. The energy was originally light energy in sunlight.

Q: (3) How did the energy get into the nutrients?

A: Plants absorbed the light energy using **chlorophyll** and transferred it into **chemical energy** in carbohydrates such as glucose, by photosynthesis. Plants use this glucose to make their other nutrients. Animals take in the energy when they eat plants, or eat other animals that have eaten plants.

Nutrition — Why Plants Are Different from Animals

All living organisms need to take many different substances — **nutrients** — into their bodies. Some are used to make new parts or to repair old parts. Others are used to release energy. Taking in useful substances is called **nutrition**.

In Chapter B3 you met the important groups of biological molecules: **proteins, fats and carbohydrates**. These are all **organic substances**.

Animals	Plants
Cannot make organic substances themselves	Can make organic nutrients from inorganic substances
Must take in organic substances as nutrients when they feed	Use carbon dioxide (from the air), water and mineral ions (from the soil)
Depend on plants, directly or indirectly, for food and energy	Make carbohydrates by photosynthesis, which means "making with light"

Like all reactions in organisms, photosynthesis depends on **enzymes** to help it take place.

The Photosynthesis Equation

In photosynthesis, **water and carbon dioxide react together to produce glucose and oxygen**. They will only do this in the presence of **sunlight** and **chlorophyll**.

- **Word equation:** carbon dioxide + water → glucose + oxygen (with "sunlight" written above the arrow and "chlorophyll" below it)
- **Balanced chemical equation:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

A balanced chemical equation shows the chemical formula of each reactant and product, and how many molecules of each are involved.

What the formulae tell us:

- **CO_2** (carbon dioxide): one molecule contains 1 atom of carbon (C) and 2 atoms of oxygen (O)
- **H_2O** (water): one molecule contains 2 atoms of hydrogen (H) and 1 atom of oxygen
- **$C_6H_{12}O_6$** (glucose): one molecule contains 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms
- **O_2** (oxygen): one molecule contains 2 oxygen atoms

Because a glucose molecule contains 6 carbon atoms, **six carbon dioxide molecules are needed to make one glucose molecule**. There is the same number of each kind of atom on each side of the equation:

Atom	Reactants: $6CO_2 + 6H_2O$	Products: $C_6H_{12}O_6 + 6O_2$
Carbon (C)	6 × 1 = 6	6
Hydrogen (H)	6 × 2 = 12	12

Oxygen (O)	(6 × 2) + (6 × 1) = 18	6 + (6 × 2) = 18

	Reactants (raw materials)	Products
Substances	Carbon dioxide and water	Glucose and oxygen
Where they come from or go	Carbon dioxide diffuses in from the air; water comes from the soil	Glucose is used or stored by the plant; oxygen is released into the air (or into the water, for an aquatic plant)

Remember: Light and chlorophyll are **not** reactants. Light provides the energy and chlorophyll captures it, so they are written on the arrow, never after a "+" sign.

Science in Context B5.01: Using Solar Energy to Make Fuels

- As the human population grows, we use more and more fuel for homes, industries and vehicles. Renewable energy sources such as wind and solar power are being used more, but a lot of our energy still comes from **fossil fuels**.
- Burning fossil fuels produces **carbon dioxide**, so the quantity of carbon dioxide in the atmosphere is increasing. This contributes to **global warming**, so we urgently need other ways of providing energy.
- Plants use energy from sunlight to make carbohydrates, which provide energy for their cells. This way of providing energy does not produce carbon dioxide — it **uses it up**.
- In many parts of the world, plants are grown to provide **biofuels**, which are burnt to produce electricity or to move vehicles. But this takes up land that may be needed for food crops, or that would be better left as natural forests or other habitats for wildlife.
- Scientists are researching **artificial photosynthesis** to make fuels. Chlorophyll captures energy from sunlight and helps to transfer it into carbohydrates, so researchers are looking for other substances that can do the same job. One laboratory is using **tiny particles of gold** as a substitute for chlorophyll. It works, but only very slowly. Researchers think it will be at least 2030 before artificial photosynthesis can produce significant quantities of fuel.

Q: How might the large-scale use of artificial photosynthesis help to slow down climate change?

A: Artificial photosynthesis would use energy from sunlight to make fuels from carbon dioxide, so carbon dioxide would be removed from the atmosphere as the fuel is made. When the fuel is burned, it only releases the carbon dioxide that was taken in to make it, so there is no overall increase in atmospheric carbon dioxide. Fossil fuels, by contrast, release carbon that was locked away underground for millions of years. Replacing fossil fuels with these fuels would reduce the build-up of carbon dioxide (a greenhouse gas), slowing global warming and climate change. Unlike biofuel crops, it would not need large areas of land, so forests, which absorb carbon dioxide, could be left standing.

Activity B5.01: Modelling Photosynthesis

Work in a group of four or five. In this model, **small balls represent atoms** and **pieces of egg carton represent molecules**.

You will need: egg cartons cut into molecule-shaped pieces (six each of three different shapes, Figure B5.01a–c); one complete egg carton that can hold 12 eggs, labelled to hold a glucose molecule (Figure B5.01d); 36 small balls that fit into the egg cartons (for example, table tennis balls); red, black and blue pens.

Method:

1. Use the red pen to label 18 of the balls "O" (oxygen), the black pen to label 6 of them "C" (carbon), and the blue pen to label the other 12 "H" (hydrogen).
2. Take one ball representing oxygen and two balls representing hydrogen, and put them into an angled box to make a water molecule (Figure B5.02).
3. Make five more water molecules.

4. Use the remaining balls to make six carbon dioxide molecules, choosing the correct egg carton shapes.
5. Use your water molecules and carbon dioxide molecules to make a glucose molecule, by putting the balls into the correct holes in the base and lid of the 12-hole egg carton. Use oxygen from the water only — do not use oxygen from the carbon dioxide.
6. Use your leftover balls to make oxygen molecules.
7. If you have time, take apart your glucose and oxygen molecules and turn them back into carbon dioxide and water.

How the numbers work: six water molecules use 12 H and 6 O; six carbon dioxide molecules use 6 C and 12 O. The glucose molecule takes 6 C, 12 H and 6 O. The 12 O left over make six O_2 molecules. All 36 balls are used both before and after the "reaction" — the atoms are only rearranged.

Q: (1) Explain how this model represents the balanced equation for photosynthesis.

A: Each ball is an atom and each carton shape is a molecule. You start with six carbon dioxide molecules (6 C and 12 O) and six water molecules (12 H and 6 O), which match the reactants $6CO_2$ + $6H_2O$. You rearrange exactly the same 36 balls into one glucose molecule (6 C, 12 H, 6 O) and six oxygen molecules (12 O), which match the products $C_6H_{12}O_6$ + $6O_2$. No balls are added or lost, which shows that there is the same number of each type of atom on both sides — the equation is balanced. The numbers of each type of carton show the numbers in front of each formula.

Q: (2) Suggest how you could modify the model, or the way that you use the model, to show how energy in sunlight helps the reaction to take place.

A: For example, only allow the molecules to be taken apart and rebuilt while a torch or lamp ("sunlight") is shining on the group, and only by a group member holding a green card representing chlorophyll. That person could collect "energy tokens" from the lamp and hand one over for each bond that is broken and remade, then place the tokens inside the glucose carton. This shows that chlorophyll captures light energy and that the energy ends up stored in glucose.

Q: (3) You may hear people say that "photosynthesis changes carbon dioxide into oxygen". Use your model to explain why this is not correct.

A: In the model, no ball ever changes into a different kind of ball — a carbon atom stays a carbon atom. The carbon dioxide molecules are broken apart and their atoms are rearranged: the carbon atoms from carbon dioxide end up in the glucose molecule, not in oxygen. Water is also a reactant, and its hydrogen atoms end up in glucose too. So photosynthesis turns carbon dioxide **and water** into **glucose and oxygen**; carbon dioxide is not simply "changed into" oxygen. (Beyond the syllabus: experiments with labelled atoms show that, in real plants, the oxygen gas released actually comes from the water molecules, so the model is a simplification.)

Reflection: Some people find that handling objects and moving them around, as in this model, helps them to understand. Think about whether the model helped you to understand the photosynthesis equation and how — for example, counting the balls shows clearly why the numbers must balance. If it did not help, suggest another method that might suit you better, such as drawing the atoms as coloured circles, building molecules from modelling clay, or watching an animation.

Chlorophyll — Capturing Energy from Sunlight

In the chloroplasts of a plant's leaves, water and carbon dioxide react together to make carbohydrates and oxygen. In other circumstances, you could shine light on a mixture of carbon dioxide and water for ever, and no carbohydrates or oxygen would ever appear. What is special about a plant's leaves is that they contain a green pigment called **chlorophyll**, which is present in their **chloroplasts**.

How chlorophyll works, step by step:

1. Chlorophyll **captures energy from sunlight** (Figure B5.03).
2. It immediately **passes the energy on (transfers it)** to water molecules and carbon dioxide molecules.

3. The energy makes these substances **react**, producing a kind of carbohydrate called **glucose** (and oxygen).
4. The glucose made in the leaves contains some of the energy that was originally in the sunlight.

So chlorophyll **transfers energy from light into energy in carbohydrates** (light energy becomes chemical energy). Figure B5.03 shows a forest described as "a giant carbohydrate factory", using sunlight as fuel and carbon dioxide and water as its raw materials.

How a Plant Uses Carbohydrates

Photosynthesis has two products: **glucose and oxygen**.

- The **oxygen** is released into the atmosphere (or into the water, if the plant is aquatic).
- The **glucose** (a carbohydrate) is used for many different purposes, described below.

Releasing Useful Energy

Some of the glucose is used to provide **energy** for activities that the plant's cells need to carry out. For example:

- Energy is needed to move **mineral ions into the root hairs by active transport**.
- Energy is needed to **build protein molecules from amino acids**, for growth.

The energy is released from glucose by **respiration** (Chapter B9).

Storing, to Use Later — Starch

Plants usually make much more glucose than they need to use for energy immediately. They store it by turning it into **starch**.

- Starch molecules are made by linking many glucose molecules together into a **long, spiral chain**.
- Millions of these molecules clump together to make **starch grains**. Figure B5.04 is a photomicrograph of potato tuber cells; iodine solution has been added, so the starch grains are stained blue.

Why starch is a good storage substance:

- Starch molecules are **not soluble in water**, so they do not get involved in the chemical reactions taking place inside the cells.
- They **do not affect the concentration of the solutions** inside the cell, so they do not cause water to enter or leave by **osmosis**.
- They can be **quickly and easily broken down to glucose** again when the plant needs it.

Making Sucrose, for Transport

- Plants can only make glucose in the parts that contain chlorophyll, which usually means the **leaves**. All the other parts of the plant must have carbohydrate delivered to them.
- Plants do not actually transport glucose. They first change it into a sugar with larger molecules, called **sucrose**.
- Sucrose is carried from one part of the plant to another inside **phloem tubes** (Chapter B7).
- When the sucrose reaches its destination, it can be changed back to glucose and used in **respiration** to release energy.

Making Cellulose, to Build Cell Walls

- As plants grow, they make new cells. Every cell needs a **cell wall**, so a growing plant must make **cellulose** molecules to form these walls.
- Cellulose is made by linking glucose molecules in long chains. The molecules are linked in a different way from starch, so the chains **stay straight** rather than coiling up into spirals.

Making Nectar, to Attract Pollinators

- Many plants reproduce sexually, producing male and female gametes in flowers.

- Unlike animal gametes, the male gametes of flowers **cannot move themselves** from place to place. Instead, they rely on **insects, bats or birds** to carry them, inside **pollen grains**, from one flower to another.
- No animal will do this unless it gets a reward, so flowers produce **nectar** that animals can feed on. Nectar contains different kinds of sugar, all made from the glucose that the plant has made by photosynthesis.
- Figure B5.05 shows insects visiting flowers to collect carbohydrate-rich nectar and protein-rich pollen. Without knowing it, they help the flower by pollinating it.

Making Amino Acids, to Make Proteins

- Plants use some of the glucose made in photosynthesis to make **amino acids**, which are then used to make **proteins**, for growth.
- Proteins contain carbon, hydrogen and oxygen, **and also nitrogen**. So to make amino acids from glucose, plants need a source of nitrogen.
- They get nitrogen from the soil in the form of **nitrate ions**. These are usually taken in by **active transport** through the root hairs, then transported to all parts of the plant, where they are combined with glucose to make amino acids.
- If a plant cannot get enough nitrate ions, it cannot synthesise proteins effectively, so it will **not grow quickly or strongly**.

Making Other Substances, such as Chlorophyll

- Glucose can also be used to make **chlorophyll**.
- Chlorophyll is **not a protein**, but it does contain **nitrogen**. It also contains **magnesium**. Therefore plants need to take in **magnesium ions**, as well as nitrate ions, to make chlorophyll.
- Without these ions, the plant's leaves look **yellow** rather than green. Figure B5.06 shows leaves on an orange tree growing in soil that does not contain enough magnesium ions; the tree cannot make enough chlorophyll, so the leaves are not completely green.
- If a plant cannot make lots of chlorophyll, it cannot photosynthesise well, and so it will not grow well.

Summary Table: Uses of Glucose

What the glucose is used for	Purpose	Key detail
Respiration	Releases useful energy	Energy for active transport of mineral ions and for building proteins
Making starch	Energy storage	Insoluble; does not affect osmosis; easily broken back down to glucose
Making sucrose	Transport	Carried in phloem tubes to parts that cannot photosynthesise
Making cellulose	Building cell walls	Straight chains of glucose; every new cell needs a wall
Making nectar	Attracting pollinators	A sugary reward for insects, bats or birds that carry pollen
Making amino acids, then proteins	Growth	Needs nitrogen from nitrate ions
Making chlorophyll	Absorbing light energy	Needs nitrogen (from nitrate ions) and magnesium (from magnesium ions)

Mineral Ions: Nitrate and Magnesium

	Nitrate ions	Magnesium ions
Element supplied	Nitrogen	Magnesium
Why needed	To make amino acids, and then proteins (nitrogen is also needed for chlorophyll)	To make chlorophyll
Deficiency symptoms	Weak growth, yellow leaves	Yellowing of leaves, often especially between the veins

Both deficiencies cause yellow leaves because both nitrogen and magnesium are needed to make chlorophyll. A plant short of either ion photosynthesises less and grows poorly.

B5.02 Leaves

A Leaf Is a Carbohydrate Factory

Photosynthesis happens inside **chloroplasts**. This is where the **enzymes** and **chlorophyll** that catalyse and supply energy for the reaction are found. In a typical plant, most chloroplasts are in the cells in the leaves, so a leaf is a factory for making carbohydrates. It is not surprising that most plants have leaves that are perfectly adapted to help photosynthesis take place as quickly and efficiently as possible.

The Structure of a Leaf

Figure B5.07a shows the parts of a leaf:

- **Lamina** — the broad, flat part of the leaf.
- **Leaf stalk** — joins the lamina to the rest of the plant. Inside the leaf stalk are collections of parallel tubes called **vascular bundles**.
- **Midrib** — the main vein running down the middle of the lamina.
- **Veins** — the vascular bundles branching through the lamina. The tubes in the vascular bundles carry substances to and from the leaf (Chapter B7).

Figure B5.07a also shows where a transverse section of the leaf and a cross-section of a vein are cut. Figure B5.07b shows the **network of veins** in a dicot leaf, which carries water to all parts of it.

Inside the leaf cells, photosynthesis happens in chloroplasts. The chlorophyll is **spread out on membranes** inside the chloroplasts, so that a lot of sunlight can reach it.

Why Most Leaves Are Broad and Thin

The raw materials for photosynthesis are carbon dioxide and water, and sunlight is needed to provide energy. Most — but not all — leaves have a **large surface area** and are **very thin**.

Feature	How it helps photosynthesis
Large surface area	Allows large amounts of sunlight to fall onto the leaf; also increases the rate at which carbon dioxide can diffuse into the leaf from the air
Thin	Sunlight can pass right through the leaf, so many cells inside it can photosynthesise; carbon dioxide can reach all the cells quite quickly by diffusion
Network of veins	Xylem vessels in the veins bring water from the soil to all parts of the leaf

Only **0.04%** of the air is carbon dioxide, so the structure of a leaf must allow carbon dioxide to move into the leaf really easily.

Water is brought to the leaf from the soil in tubes called **xylem vessels** (pronounced "zi-lem"). These run in the vascular bundles, which form the veins in the leaf. You can often see many tiny veins if you look closely at a leaf.

How the Raw Materials Reach a Palisade Cell

Figure B5.08 summarises how the raw materials for photosynthesis travel to a chloroplast in a palisade cell:

What is needed	How it reaches the chloroplasts
Sunlight (energy)	Passes through the transparent cuticle and upper epidermis
Carbon dioxide	Diffuses through the stomata from the atmosphere, then diffuses through the air spaces to the palisade cells
Water	Travels from the roots in xylem vessels, then travels to the chloroplasts by osmosis

Tissues in a Leaf

Although a leaf is thin, it is made up of several layers of cells. You can see these if you look at a **transverse section (TS)** of a leaf under a microscope (Figure B5.09). Each tissue in a leaf has its own function. From top to bottom:

Tissue or structure	What it is like	Function and adaptation
Cuticle	A waxy substance forming a thin, transparent, waterproof covering	Reduces water loss; transparent, so light passes straight through
Upper epidermis	A layer of transparent cells packed tightly together, with no chloroplasts	Reduces the quantity of water vapour escaping from the leaf; lets light through to the palisade cells
Palisade mesophyll	Tall, narrow cells containing very large numbers of chloroplasts, close to the top of the leaf	Main function is photosynthesis; being near the top, they get plenty of sunlight through the transparent epidermis above
Spongy mesophyll	Rounded, loosely packed cells with chloroplasts and many air spaces between them	Also photosynthesise; the air spaces let carbon dioxide and oxygen diffuse between the air and the cells inside the leaf, and let water vapour move from the cell surfaces to the outside of the leaf
Vascular bundle (vein)	Xylem vessels and phloem tubes side by side	Xylem brings water and mineral ions into the leaf; phloem carries sucrose away to the rest of the plant
Lower epidermis	A layer of cells that usually has no cuticle	The underside of the leaf does not often have sunlight falling on it, so it does not get as hot and does not lose as much water vapour
Stomata	Openings in the lower epidermis	Allow diffusion of carbon dioxide and oxygen in and out of the leaf; water vapour also diffuses out through

		them
Guard cells	A pair of cells around each stoma; they contain chloroplasts	Change their shape to open and close the stomata

The mesophyll layer is made up of the palisade mesophyll and spongy mesophyll together. Figure B5.09 also shows the inside of a palisade cell: a cell wall, cytoplasm, a nucleus, a large vacuole and many chloroplasts.

The Palisade Mesophyll Cell

The palisade mesophyll cell is the leaf cell that is best adapted for photosynthesis:

- It contains **very large numbers of chloroplasts**, so lots of chlorophyll is available to absorb light energy.
- It is **tall and narrow**, and the cells are packed side by side in a layer, so many cells fit into the top part of the leaf where light is brightest.
- It is **close to the upper surface**, just below the transparent cuticle and upper epidermis, so it gets plenty of sunlight.
- It is close to **air spaces** (supplying carbon dioxide by diffusion) and to the **veins** (supplying water, which moves to the chloroplasts by osmosis).

Identifying Leaf Structures in Photomicrographs

You need to identify leaf structures in diagrams and in photomicrographs of a dicotyledonous leaf, such as Figure B5.10 (part of a tea plant leaf, ×400). Tips for recognising each layer:

- **Upper epidermis** — the single row of cells forming the top surface; the cuticle is the thin line on top of it.
- **Palisade mesophyll** — the row of tall, narrow cells packed tightly side by side just below the upper epidermis; they look darker because they are full of chloroplasts.
- **Spongy mesophyll** — rounded, irregular cells with many spaces between them, in the lower part of the leaf.
- **Vascular bundle** — a cluster of cells (xylem and phloem), often near the middle of the leaf.
- **Lower epidermis** — the row of cells on the bottom surface. Look for a gap (a **stoma**) between two small **guard cells**, usually with a large air space just above it.

Summary: How the Leaf Is Adapted for Photosynthesis

Feature of the leaf	Why it is an adaptation for photosynthesis
Large surface area	Captures more sunlight; more carbon dioxide can diffuse in
Thin	Light reaches all the cells; short diffusion distance for carbon dioxide
Transparent cuticle and upper epidermis	Let light through to the palisade layer
Palisade cells near the top, packed with chloroplasts	Absorb as much light energy as possible
Spongy mesophyll with air spaces	Carbon dioxide can diffuse to all the photosynthesising cells
Stomata with guard cells	Let carbon dioxide in and oxygen out; can be closed to reduce water loss

Network of veins (xylem and phloem)	Xylem supplies water; phloem carries away the sucrose made
Chlorophyll spread out on membranes in chloroplasts	A lot of sunlight can reach it

B5.03 Factors Affecting Photosynthesis

What Affects the Rate of Photosynthesis?

The **rate** of photosynthesis is how fast it happens — for example, how much oxygen is given off in unit time. The rate is affected by several **environmental factors**:

- **The supply of the raw materials** — carbon dioxide and water
- **The quantity of sunlight**, which provides energy for the reactions
- **The temperature**, because this affects the activity of enzymes

As well as these environmental factors, the **quantity of chlorophyll** in the leaf also helps to determine how fast photosynthesis can take place.

Factor	Effect of increasing it	Reason
Light intensity	Rate increases, until another factor limits it	More energy is available for the reactions
Carbon dioxide concentration	Rate increases, until another factor limits it	More of a raw material is available
Temperature	Rate increases up to an optimum, then falls quickly	Enzymes work faster as temperature rises, but are denatured at high temperatures
Quantity of chlorophyll	More chlorophyll allows a faster rate	More light energy can be absorbed

Limiting Factors

Suppose a plant has plenty of light and a warm temperature, but very little carbon dioxide. Making the light brighter will not make it photosynthesise any faster, because the shortage of carbon dioxide is holding the rate back. The factor that is in shortest supply, and is holding back the rate, is called the **limiting factor**.

- On a graph of rate against a factor, the line **rises** while that factor is limiting, then **levels off (plateaus)** when some other factor becomes limiting.
- This explains why glasshouse growers add extra heat, light or carbon dioxide (Project B5.01), and why the students in Practice Question 6 put their syringes in **bright sunshine**: so that light would not limit the rate and they could see the effect of carbon dioxide.

Experimental Skills B5.01: Testing a Leaf for Starch

Aim: To find out whether a leaf contains starch. Leaves turn some of the glucose that they make by photosynthesis into starch, so if we find starch in a leaf, that tells us it has been photosynthesising. You also practise working safely, making careful observations and drawing conclusions.

You will need: a 250 cm^3 beaker; a burner, tripod and gauze; a boiling tube; some iodine solution, with a dropper; forceps; a glass rod; some alcohol (methylated spirits) — collect this only when you are ready to use it; a fresh green leaf that has been photosynthesising; a white tile.

Safety:

- Alcohol is flammable (burns easily).

- Do not collect the alcohol until you have turned off your burner.
- If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method (Figure B5.11):

1. Boil some water in a beaker. Take a green leaf from a healthy plant and drop it into the boiling water. This breaks down the cell membranes in the leaf. Leave the leaf in the boiling water for about 30 seconds.
2. Turn off the flame.
3. Put some alcohol into a boiling tube. Stand the tube of alcohol in the hot water.
4. Use the forceps to remove the softened leaf from the hot water and drop it into the tube of alcohol. Chlorophyll is soluble in alcohol, so it comes out of the leaf. Leave the leaf in the alcohol until all the chlorophyll has come out.
5. The leaf will now be brittle (it will break easily). Carefully remove it from the alcohol, and dip it into hot water again to soften it.
6. Spread out the leaf on a white tile and cover it with iodine solution. A blue-black colour shows that the leaf contains starch.

Step	Why it is done
Boil the leaf in water	Breaks down the cell membranes so that iodine solution can reach the starch inside the cells; it also kills the cells
Turn off the flame before using alcohol	Alcohol is flammable and must never be near a naked flame
Stand the tube of alcohol in hot water	Warms the alcohol safely so it dissolves the chlorophyll faster
Leave the leaf in alcohol	Removes the green chlorophyll, so the leaf goes pale and colour changes are easy to see
Dip the leaf in hot water again	Softens the brittle leaf so it can be spread out without breaking
Use a white tile	Makes the colour of the leaf easy to see
Add iodine solution	Turns blue-black where starch is present; stays orange-brown where there is none

Variables: Nothing is varied — this is a standard test that you will use in Experimental skills B5.02, B5.03 and B5.04.

Expected results: A leaf from a plant that has been photosynthesising turns **blue-black**. Any part without starch stays the **orange-brown** colour of the iodine solution.

Conclusion: A blue-black leaf contains starch, so it has been photosynthesising.

Q: (1) Describe the results of your experiment. What conclusion can you make?

A: When iodine solution was added, the pale leaf turned blue-black. (Iodine solution itself is orange-brown, and it would have stayed orange-brown if there was no starch.) Conclusion: the leaf contained starch, so it had been photosynthesising.

Q: (2) Cell membranes are partially permeable. They will not allow starch or iodine molecules to pass through them. Use this information to explain why it was necessary to boil the leaf.

A: The starch is inside the cells, surrounded by cell membranes. Iodine molecules cannot pass through intact cell membranes, so the iodine solution could not reach the starch and no colour change would be seen. Boiling the leaf breaks down the cell membranes, so the iodine solution can get into the cells and react with the starch.

Q: (3) Suggest why it is a good idea to remove the chlorophyll from the leaf, before adding iodine solution to it.

A: The dark green colour of chlorophyll would hide (mask) the colour of the iodine solution, so it would be difficult to tell whether the leaf had turned blue-black. Removing the chlorophyll leaves the leaf pale, so the colour change is easy to see.

Experimental Skills B5.02: Investigating the Necessity for Chlorophyll for Photosynthesis

Aim: To find out whether chlorophyll is needed for photosynthesis, using the starch test on a **variegated leaf** — a leaf that has some white areas and some green areas (Figure B5.12). You also practise recording observations with a labelled drawing.

You will need: the same apparatus as for Experimental skills B5.01 Testing a leaf for starch; a fresh variegated leaf, taken from a plant that has been photosynthesising.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method:

1. Make a drawing of your variegated leaf. Do not colour it. Label the parts that are green and the parts that are white.
2. Test the leaf for starch, following the procedure described in Experimental skills B5.01.
3. Record your results by drawing and labelling the leaf after you have added iodine solution to it.

Variable or feature	In this investigation
Independent variable	Whether chlorophyll is present (green areas) or absent (white areas)
Dependent variable	Whether starch is present (colour with iodine solution)
Controlled variables	Both areas are on the same leaf, so they had the same light, carbon dioxide, water, temperature and time
Control	The green parts of the leaf

Expected results: The areas that were green turn **blue-black**; the areas that were white stay **orange-brown**. The pattern of blue-black on the second drawing matches the pattern of green on the first drawing.

Conclusion: Only the parts of the leaf that contain chlorophyll made starch, so chlorophyll is necessary for photosynthesis.

Q: (1) Which parts of the leaf contained starch?

A: Only the parts that were green (containing chlorophyll) — they turned blue-black. The white parts stayed orange-brown, so they contained no starch.

Q: (2) Write a brief conclusion for your experiment.

A: Starch was only made in the green parts of the leaf, which contain chlorophyll. The white parts, which have no chlorophyll, did not make starch. This shows that chlorophyll is necessary for photosynthesis.

Experimental Skills B5.03: Investigating the Necessity for Light for Photosynthesis

Aim: To use the starch test to compare the starch content of parts of a leaf that have received light with parts that have not. You also practise recording results with a labelled drawing and writing a short conclusion.

You will need: a healthy plant growing in a pot, which has been in a dark cupboard for two days; some black paper, scissors and paper clips; (after one day) the same apparatus as for Experimental skills B5.01 Testing a leaf for starch.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately.

Method:

1. Cut out a shape from the black paper, which you can attach to a leaf. If you like, you can cut your initials into the paper (Figure B5.13).
2. Attach the paper to a leaf on the potted plant. Do not take the leaf off the plant!
3. Put the plant into a place where it gets plenty of light. Leave it until the next day.
4. Remove the leaf from the plant. Remove the black paper from the leaf.
5. Test the leaf for starch.
6. Record your results by drawing and labelling the leaf after you have added iodine solution to it.

Destarching: Before the experiment, the plant was kept in a dark cupboard for two days. With no light it could not photosynthesise, so it used up its starch stores. **Destarching** means leaving a plant in the dark for long enough for it to use up its starch stores.

Variable or feature	In this investigation
Independent variable	Whether the part of the leaf received light (uncovered) or not (covered by black paper)
Dependent variable	Whether starch is present (colour with iodine solution)
Controlled variables	Same leaf on the same plant, so the same chlorophyll, carbon dioxide, water, temperature and time
Control	The part of the leaf that was not covered

Expected results: The part covered by black paper stays **orange-brown** (no starch). The uncovered part turns **blue-black** (starch). If you cut your initials into the paper, light shone through the holes, so your initials show up blue-black on an orange-brown background.

Conclusion: Only the parts of the leaf that received light made starch, so light is necessary for photosynthesis.

Why only part of the leaf is covered: If you had covered all of the leaf and found that none of it had photosynthesised, you would not be sure that the reason was because it had no light. It could have been because the whole plant was not photosynthesising.

Q: (1) The potted plant was put into a cupboard to make sure it used up all its starch stores and had no light to be able to make more. This is called destarching. Suggest why this was a necessary part of this experiment.

A: Destarching makes sure there is no starch in the leaf at the start. Then any starch found at the end must have been made by photosynthesis during the experiment, in the parts that received light. If the leaf still contained old starch, the covered part might test positive even though it had not photosynthesised, and you could not tell whether light is needed.

Q: (2) Explain why, in this experiment, the leaf had to remain attached to the plant for a day, after you put the black paper onto it.

A: The leaf needed time in the light to photosynthesise and make enough starch to be detected by the iodine test. It had to stay attached to the plant so that it stayed alive and healthy and kept receiving water (through the xylem from the roots), which is a raw material for photosynthesis. A leaf removed from the plant would wilt and would not photosynthesise normally.

Q: (3) Use your results to write a short conclusion to your experiment.

A: The part of the leaf covered by black paper did not contain starch (it stayed orange-brown), but the uncovered part did (it turned blue-black). This shows that light is necessary for photosynthesis.

Controls

A **control** is a standard sample that you use as a comparison, to find the effect of changing a variable. The control is the same as the test sample in every way except for the one factor being investigated.

- In Experimental skills B5.02, you wanted to find out whether chlorophyll is needed for photosynthesis. The white parts of the leaf had no chlorophyll, and you could compare them with the normal, green parts. The **green parts were the control**.
- In Experimental skills B5.03, you wanted to find out whether light is needed. You needed to compare a part of a leaf that did have light with a part that did not. The **part of the leaf that you did not cover was the control**. It allowed you to compare the part that did not have light with this normal part, which did.

Experimental Skills B5.04: Investigating the Necessity for Carbon Dioxide for Photosynthesis

Aim: To compare the ability of two leaves on the same plant — one with carbon dioxide and one without — to make starch. This is not an easy experiment to set up, so you will probably use one set of apparatus for the whole class.

You will need: a healthy plant growing in a pot, which has been destarched (see Experimental skills B5.03); two transparent containers, e.g. conical flasks; two rubber bungs to fit the flasks, each with a slit in one edge in which a leaf stalk can fit; petroleum jelly; a method of supporting the transparent containers; some potassium hydroxide solution; some distilled water; (after one day) the same apparatus as for Experimental skills B5.01 Testing a leaf for starch.

Safety: Alcohol is flammable (burns easily). Do not collect the alcohol until you have turned off your burner. If your alcohol (or someone else's) does start to burn, put a damp cloth over it immediately. Potassium hydroxide solution is also corrosive, so wear eye protection and avoid skin contact.

Method (Figure B5.14):

1. Set up the apparatus as shown in Figure B5.14. Find a way of supporting the flasks so that they do not pull on the leaves — you could use a retort stand and clamps. Put potassium hydroxide solution into one flask and distilled water into the other. Hold one of the leaf stalks carefully, slot it into the slit in one of the rubber bungs, then gently push the flask over the bung and into its support. Repeat with the other leaf and flask.
2. Rub petroleum jelly over the rubber bungs, to try to prevent any air entering or leaving the flasks (an airtight seal).
3. Place the apparatus in a light place and leave it for at least one day.
4. Test both leaves for starch.
5. Record your results.

Figure B5.14 shows a potted plant with two of its leaves (still attached) each sealed inside a flask: one flask contains potassium hydroxide solution to absorb carbon dioxide, and the other contains distilled water.

Variable or feature	In this investigation
Independent variable	Whether carbon dioxide is present (distilled water flask) or absent (potassium hydroxide flask)

Dependent variable	Whether starch is present in each leaf (colour with iodine solution)
Controlled variables	Same plant, same light, same temperature, same time; both leaves sealed in a flask containing a liquid
Control	The leaf in the flask containing distilled water

Expected results: The leaf in the flask with potassium hydroxide solution stays **orange-brown** (no starch). The leaf in the flask with distilled water turns **blue-black** (starch).

Conclusion: The leaf without carbon dioxide could not make starch, so carbon dioxide is necessary for photosynthesis.

Q: (1) What was the control in this experiment?

A: The leaf in the flask containing distilled water. It had carbon dioxide in the air around it, but was otherwise treated in exactly the same way as the leaf in the potassium hydroxide flask.

Q: (2) Suggest why distilled water was added to one flask, rather than just leaving it empty.

A: So that the two flasks were the same in every way except for the presence of carbon dioxide. Potassium hydroxide solution is mostly water, so it makes the air in its flask humid. Putting the same volume of distilled water (which does not absorb carbon dioxide) in the other flask keeps the conditions, such as humidity, the same. Then any difference between the two leaves can only be caused by the carbon dioxide.

Q: (3) Write a short conclusion for your experiment.

A: The leaf in the flask with potassium hydroxide solution (no carbon dioxide) did not contain starch, while the leaf in the flask with distilled water (carbon dioxide present) did contain starch. This shows that carbon dioxide is necessary for photosynthesis.

Comparing the Three "Necessity" Investigations

Investigation	Factor removed	How it is removed	Control	Result in the test part	Conclusion
B5.02	Chlorophyll	Variegated leaf: the white areas have none	Green areas	No starch in the white areas	Chlorophyll is needed
B5.03	Light	Black paper covers part of the leaf	Uncovered part of the leaf	No starch under the paper	Light is needed
B5.04	Carbon dioxide	Potassium hydroxide solution absorbs it	Leaf in the flask of distilled water	No starch in the leaf with potassium hydroxide	Carbon dioxide is needed

In B5.03 and B5.04 the plant is **destarched** first, so that any starch found must have been made during the experiment.

Experimental Skills B5.05: Investigating the Effect of Varying Light Intensity on the Rate of Photosynthesis

Aim: To find out how light intensity affects the rate of photosynthesis, using an **aquatic plant** (a plant that lives in water). An aquatic plant makes it easy to measure the rate, because you can see **bubbles of oxygen** being released into the water. To measure a rate, you need to measure how much oxygen is given off in **unit time** — a definite length of time, such as one minute.

You will need: a healthy aquatic plant, which has been in the light and has been photosynthesising; the apparatus shown in Figure B5.15; a paperclip; a lamp; a ruler to measure the distance between the lamp and the plant; a stopwatch or timer.

Safety: Keep water away from the lamp.

Before you start: Check that you know how to use your stopwatch or timer, and that it is working properly.

Method:

1. Collect a piece of aquatic plant. Cut off a piece about 7 to 9 cm long. Work out which way up it is. Attach a paperclip to the top of it.
2. Set up the apparatus as shown in Figure B5.15. The paperclip will hold the top of the plant down, so that its cut end floats upwards.
3. Place the lamp as close as possible to the beaker. Measure the distance, and record it in a results table.
4. Start the stopwatch or timer. Count the number of bubbles released from the cut end of the stem in one minute. Record this, and then repeat two more times.
5. Move the lamp a little bit further from the plant. Measure the new distance, and repeat step 4.
6. Repeat step 5 several times, until you have bubble counts for at least five different distances. Do more if you have time.
7. Plot your results as a line graph, with distance on the x-axis and mean number of bubbles on the y-axis. Make sure that the two axes are labelled fully, and that you choose suitable scales. Join the points using ruled, straight lines between the points.

Figure B5.15 shows a lamp on a stand at one end of a metre rule, a transparent heat shield, and a beaker of water containing the cut stem of the aquatic plant, with bubbles containing oxygen produced in photosynthesis rising from the cut end.

The results table in the book has these columns (the last four are all under the heading "Number of bubbles produced in one minute"):

Distance of lamp from beaker / cm	1st try	2nd try	3rd try	Mean

To work out each mean, add the three counts and divide by 3. For example, counts of 24, 26 and 25 bubbles give a mean of (24 + 26 + 25) ÷ 3 = 25 bubbles per minute.

Variable or feature	In this investigation
Independent variable	Light intensity, changed by moving the lamp to different distances from the beaker
Dependent variable	Rate of photosynthesis, measured as the number of bubbles released from the cut stem in one minute
Controlled variables	Temperature (kept steady by the heat shield); carbon dioxide concentration in the water; the same piece of plant; the same lamp; the same counting time; other light in the room kept low
Repeats	Three counts at each distance, then a mean is calculated

Expected results: The closer the lamp, the more bubbles are produced per minute. As the lamp is moved further away, the number of bubbles per minute decreases, so the line on the graph slopes downwards from left to right.

Conclusion: Light intensity decreases as the lamp is moved further away, so the results show that **the higher the light intensity, the faster the rate of photosynthesis**.

Q: (1) Explain why an aquatic plant was used in this experiment.

A: An aquatic plant releases the oxygen made in photosynthesis into the water as visible bubbles, which can be counted (or collected). This makes it easy to measure how much oxygen is given off in unit time, which is the rate of photosynthesis. A land plant releases oxygen into the air, where it cannot be seen or easily measured.

Q: (2) Light intensity decreases as the distance of the lamp from the beaker increases. Write a conclusion to summarise what your results show about the effect of light intensity on the rate of photosynthesis.

A: As the lamp was moved further from the beaker, the mean number of bubbles produced per minute decreased. Because light intensity decreases with distance, this shows that the higher the light intensity, the faster the rate of photosynthesis. Light provides the energy for photosynthesis, so more light energy allows the reaction to happen faster.

Q: (3) Lamps produce heat as well as light. Explain why you would be unable to draw a conclusion about light intensity, if you did not use a heat shield.

A: Without a heat shield, the lamp would warm the water more when it was close to the beaker than when it was far away. Temperature also affects the rate of photosynthesis, because it changes the activity of enzymes. So light intensity and temperature would both be changing at the same time, and you could not tell whether a change in the number of bubbles was caused by the light or by the heat. The transparent heat shield lets light through but stops most of the heat reaching the plant, so light intensity is the only variable that changes.

Q: (4) It is possible to collect the gas given off by the plant and measure its volume. Suggest why this would be an improvement on the method you have used here.

A: Bubbles are not all the same size, so counting them does not measure the amount of oxygen accurately. Bubbles may also come off too fast to count, and very small bubbles may be missed. Collecting the gas (for example in an upturned measuring cylinder, a gas syringe or a capillary tube) and measuring its volume in cm^3 gives a more accurate measurement of how much gas is produced per minute.

Self Assessment: Drawing a Good Line Graph

After drawing your graph, give yourself a mark for each feature below: **2 marks** if you did it really well; **1 mark** if you made a good attempt at it and partly succeeded; **0 marks** if you did not try to do it, or did not succeed.

- The axes are both fully labelled, including units.
- The scales on both axes go up in even steps.
- The scales use at least half of the graph paper.
- The points are plotted as small, neat crosses in exactly the right place.
- The lines between the points are drawn with a ruler, and exactly meet the centres of the crosses.

How many marks do you give yourself out of 10? What will you do next time you draw a line graph, to increase your score?

Experimental Skills B5.06: Investigating the Effect of Temperature and Carbon Dioxide on the Rate of Photosynthesis

Aim: To plan and carry out your own experiment, using a method similar to Experimental skills B5.05, but with **either carbon dioxide concentration or temperature** as the independent variable.

You will need: the apparatus used in Experimental skills B5.05; electric water-baths, or access to ice and hot water, and a thermometer.

Method:

1. Think about how you will vary temperature. You can stand the beaker in a water-bath at different temperatures, or add ice and hot water to the water around the plant. Aim to have at least five values

of your independent variable.
2. Think about other variables that you should try to keep constant, and how you will do this.
3. Write out your plan. You could structure it like this: the independent variable and how you will vary it; the values of the independent variable you will use; the variables that you will keep constant, and how you will do this; the variable that you will measure, and how you will do this; possible safety risks, and how you will minimise them; an outline results chart, ready to complete as you work.
4. If possible, exchange your plan with another person in your class who is investigating the other variable. Give them feedback on their plan, and use their feedback to make improvements to your own plan.
5. When you are ready, carry out your experiment and record your results.

An example plan for each version:

Plan heading	Temperature investigation	Carbon dioxide investigation
Independent variable and how to vary it	Temperature: stand the beaker in electric water-baths at different temperatures, or add ice or hot water to the water around the plant	Carbon dioxide concentration: dissolve different amounts of sodium hydrogencarbonate in the water
Values to use (at least five)	For example 10, 20, 30, 40 and 50 °C	For example 0, 0.1, 0.2, 0.3 and 0.4% sodium hydrogencarbonate solution
Variables kept constant, and how	Light intensity (same lamp at a fixed distance, with a heat shield); carbon dioxide (same concentration of sodium hydrogencarbonate); same piece of plant; same counting time	Temperature (water-bath, checked with a thermometer); light intensity (same lamp at a fixed distance); same piece of plant; same counting time
Variable measured, and how	Number of bubbles released in one minute, counted three times to find a mean (or volume of gas collected per minute)	Same as for temperature
Safety risks, and how to minimise them	Hot water can scald, so handle it carefully; keep water away from the lamp and electrical leads; take care with glassware	Keep water away from the lamp; wipe up spills; take care with glassware
Outline results chart	Temperature / °C; number of bubbles in one minute (1st, 2nd and 3rd try); mean	Concentration of sodium hydrogencarbonate / %; number of bubbles in one minute (1st, 2nd and 3rd try); mean

Expected results:

- **Temperature:** the rate increases as the temperature rises, up to an optimum temperature, then falls quickly at higher temperatures because the enzymes that catalyse photosynthesis are denatured.
- **Carbon dioxide:** the rate increases as the carbon dioxide concentration increases, then levels off when another factor (such as light intensity) becomes the limiting factor.

Q: (1) Plot a line graph to display your results.

A: Put the independent variable on the x-axis, labelled "Temperature / °C" or "Concentration of sodium hydrogencarbonate solution / %". Put "Mean number of bubbles per minute" on the y-axis. Use even scales that fill at least half of the grid, plot each mean as a small, neat cross, and join the crosses with ruled straight lines. For temperature, the line should rise to a peak at the optimum and then fall steeply. For carbon dioxide, the line should rise and then level off.

Q: (2) Write a short conclusion.

A: Temperature: "As the temperature increased, the rate of photosynthesis increased up to an optimum temperature (read the exact value from your graph), and then decreased. This is because the enzymes that catalyse photosynthesis work faster as temperature rises, but are denatured at high temperatures." Carbon dioxide: "As the carbon dioxide concentration increased, the rate of photosynthesis increased, because more of this raw material was available. At the highest concentrations the rate levelled off, because another factor, such as light intensity, became limiting."

Q: (3) You probably encountered some difficulties or problems as you did your experiment. What were they, and how did you overcome them?

A: Typical problems and solutions:

- The water temperature drifted during counting — use a thermostatically controlled electric water-bath and check the temperature with a thermometer before and after each count.
- Bubbles came off too fast, unevenly or in different sizes — count for longer, or collect the gas and measure its volume.
- The plant took time to adjust after each change — wait a few minutes at each new temperature or concentration before starting to count.
- The lamp warmed the water — use a heat shield and keep the lamp at a fixed distance.
- Different pieces of plant gave different results — use the same piece of plant throughout.

Q: (4) If you could start again, and had lots of time available, what improvements would you make to your experiment?

A: For example: measure the volume of gas produced instead of counting bubbles (or use an oxygen sensor and data logger); test more values over a wider range, with smaller steps near the optimum; repeat each measurement at least three times, calculate means and repeat any anomalous results; control all the other variables more strictly; and give the plant a few minutes to adjust to each new condition before measuring.

Project B5.01: Designing a Glasshouse

Aim: To design a glasshouse to grow a crop in your country. The glasshouse will provide ideal conditions to help the plants photosynthesise quickly and grow fast, so that they give a **high yield** for the grower.

Method: Work in a group of three or four. You might like to assign different roles to different people (e.g. organiser, researcher, communicator). Research the questions below, then give a presentation to the rest of the class. You can make your presentation as a video, use PowerPoint, or make models and draw diagrams — or all of these.

Variables to control: temperature, light intensity, carbon dioxide concentration, water supply and mineral ions — each should be kept at a level where it does not limit the rate of photosynthesis, without costing more than the extra crop is worth.

Safety considerations: Burners used to heat the glasshouse and supply carbon dioxide must be well maintained and ventilated, because faulty burners can produce poisonous carbon monoxide. Keep lamps, heaters and electrical cables away from water sprays and irrigation.

Q: What crop will you grow in the glasshouse?

A: Choose a valuable crop that grows well under cover, for example tomatoes, cucumbers, peppers, lettuce, strawberries or flowers.

Q: What conditions does this crop need to grow well?

A: Plenty of light; warm (but not too hot) temperatures — about 20–25 °C for tomatoes during the day; enough carbon dioxide; a steady supply of water; mineral ions such as nitrate ions (for proteins) and magnesium ions (for chlorophyll); and protection from pests and diseases.

Q: How can a glasshouse provide better conditions for the crop, than if it was grown outside?

A: Glass lets light in and traps heat, so it is warmer inside. Temperature, light, carbon dioxide, water and mineral ions can all be controlled, so that none of them limits the rate of photosynthesis. The crop is also protected from wind, frost, heavy rain and many pests, and it can be grown out of season or all year round.

Q: How will you control temperature in the glasshouse? In your country, will you need to heat it, cool it, or do both at different times?

A: To heat it: use electric heaters or paraffin or gas burners (which also release carbon dioxide), and insulate it at night. To cool it: open vents and windows, use fans, shade the glass, or spray a fine water mist. A thermostat can switch these on and off to keep the temperature near the crop's optimum. A cold country mainly needs heating, a hot country mainly needs cooling, and many countries need heating at night or in winter and cooling on hot days.

Q: How will you control light in the glasshouse? Will you need to give extra light, or might you need to shade it sometimes?

A: Lamps (such as LED grow lights) can raise the light intensity on dull days and lengthen the day in winter, so photosynthesis continues for longer. In very sunny, hot places, blinds, shade netting or whitewash on the glass can reduce the light so that the plants do not overheat or lose too much water.

Q: How will you provide carbon dioxide in the glasshouse? Providing it costs money, so what might be a suitable concentration to aim for, that will give a good return for your expenditure?

A: Burn paraffin, propane or natural gas (which supplies heat as well), or release carbon dioxide from cylinders. Normal air contains about 0.04% carbon dioxide. Raising it to about 0.1% (about two and a half times normal) greatly increases the rate of photosynthesis. Above about 0.1%, the rate increases very little because another factor, such as light or temperature, becomes limiting, so the extra cost would not be repaid by a bigger crop.

Expected outcome: A good design keeps every factor close to its best level, so the crop photosynthesises quickly, grows fast and gives a high yield that is worth more than the cost of heating, lighting and carbon dioxide. Remember water (irrigation or drip-feeding) and fertilisers containing nitrate and magnesium ions too.

Key Formulas and Concepts

1. Word equation for photosynthesis: carbon dioxide + water → glucose + oxygen (in the presence of sunlight and chlorophyll)
2. Balanced chemical equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
3. The equation balances: 6 carbon, 12 hydrogen and 18 oxygen atoms on each side
4. 6 molecules of CO_2 and 6 of H_2O make 1 molecule of glucose and 6 of O_2, so number of glucose molecules = number of CO_2 molecules ÷ 6 (e.g. 12 ÷ 6 = 2)
5. Chlorophyll absorbs light energy and transfers it into chemical energy stored in glucose
6. Glucose + nitrogen (from nitrate ions) → amino acids → proteins
7. Glucose + nitrogen (from nitrate ions) + magnesium (from magnesium ions) → chlorophyll
8. Glucose is used in respiration (energy) and converted to starch (storage), sucrose (transport in phloem), cellulose (cell walls) and nectar (attracting pollinators)
9. Carbon dioxide route: atmosphere → stomata → air spaces → palisade cells (by diffusion)
10. Water route: soil → roots → xylem vessels in the veins → leaf cells (by osmosis)
11. Starch test: boil in water, then warm alcohol (water bath, no flame), then soften in hot water, then add iodine solution: blue-black = starch, orange-brown = no starch
12. Rate of photosynthesis = quantity of oxygen produced ÷ time (e.g. bubbles per minute, or cm^3 of gas per minute)
13. Mean = (1st try + 2nd try + 3rd try) ÷ 3

14. Light intensity decreases as the lamp is moved further away (useful extension: light intensity is proportional to 1 ÷ distance²)
15. Air contains about 0.04% carbon dioxide; glasshouse growers often raise it to about 0.1%
16. Rate rises with light intensity and carbon dioxide concentration until another factor limits it; rate rises with temperature up to an optimum, then falls as enzymes are denatured

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Exam Focus Priorities

1. **Learn both photosynthesis equations exactly** — Word equation: carbon dioxide + water → glucose + oxygen. Balanced equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Write light and chlorophyll on the arrow, never as reactants, and check that the atoms balance.
2. **Chlorophyll transfers energy, it does not make it** — Say that chlorophyll absorbs (traps) light energy and transfers it into chemical energy in glucose. Never write that plants "make energy".
3. **Know the starch test and the reason for every step** — Boil in water (breaks cell membranes so iodine can reach the starch), turn off the flame, warm alcohol in a water bath (removes chlorophyll so the colour can be seen), soften in hot water, add iodine solution: blue-black means starch, orange-brown means none.
4. **Explain destarching** — The plant is kept in the dark for about two days so it uses up its starch; any starch found later must have been made during the experiment.
5. **Always identify the control** — Green parts of a variegated leaf (chlorophyll), the uncovered part of the leaf (light), the leaf in the flask of distilled water (carbon dioxide). The control is identical except for the factor being tested.
6. **Name the leaf tissues in order and link each to its function** — Cuticle, upper epidermis, palisade mesophyll, spongy mesophyll with air spaces, vascular bundle (xylem and phloem), lower epidermis with stomata and guard cells.
7. **Know which cells contain chloroplasts** — Palisade mesophyll cells have the most, spongy mesophyll cells have many, guard cells have a few; other epidermis cells and xylem have none. Use this to interpret data such as the bar chart in Practice Question 4.
8. **Get the routes of the raw materials right** — Carbon dioxide diffuses in through the stomata and air spaces; water travels up the xylem and then moves into cells by osmosis. Carbon dioxide does not enter through the roots.
9. **Give a reason with each use of glucose** — Starch (insoluble store, no effect on osmosis), sucrose (transport in phloem), cellulose (cell walls), amino acids and proteins (growth, need nitrate), chlorophyll (needs nitrate and magnesium), nectar (attracts pollinators), respiration (releases energy).
10. **Do not mix up nitrate and magnesium** — Nitrate ions supply nitrogen for amino acids and proteins (deficiency: weak growth, yellow leaves); magnesium ions are needed for chlorophyll (deficiency: yellowing, especially between the veins).
11. **Plan rate experiments with clear variables** — State the independent variable (light intensity via lamp distance, temperature, or carbon dioxide concentration), the dependent variable (bubbles or gas volume per minute), the controlled variables, the heat shield, and repeats with means.
12. **Draw graphs that earn every mark** — Independent variable on the x-axis, both axes labelled with units, even scales filling at least half the grid, small neat crosses, ruled straight lines from point to point, nothing extended beyond the data.
13. **Describe, then explain, data** — Describe the trend with numbers from the table or graph (for example "increased from 5 mm to 17 mm"), then explain it using raw materials, energy from light, enzymes or limiting factors.
14. **Avoid common mistakes** — Photosynthesis does not "change carbon dioxide into oxygen" (the carbon goes into glucose); plants do not get their food from the soil (only water and mineral ions);

bubble counts are not a perfect measure of oxygen because bubbles vary in size.

Textbook Questions with Answers

Question B5.01

Explain why animals are dependent on plants for their supply of energy.

Answer: Animals cannot make organic substances (carbohydrates, fats and proteins) from inorganic substances, so they must take them in as food. Plants can make organic substances by photosynthesis: chlorophyll captures light energy from the Sun and transfers it into chemical energy stored in glucose and in the other substances made from it. Animals get this energy by eating plants, or by eating other animals that have eaten plants. So all the energy in an animal's food originally came from sunlight captured by plants — without plants, animals would have no supply of energy.

Question B5.02

Explain how chlorophyll helps photosynthesis to happen.

Answer: Chlorophyll is a green pigment found in chloroplasts. It captures (absorbs) energy from sunlight and immediately transfers this energy to water molecules and carbon dioxide molecules. This energy makes the carbon dioxide and water react together to produce glucose (and oxygen). In this way, chlorophyll transfers energy from light into chemical energy in carbohydrates. Without chlorophyll, light shining on carbon dioxide and water would never produce glucose.

Question B5.03

How many glucose molecules can be made from 12 molecules of carbon dioxide and 12 molecules of water?

Answer: 2 glucose molecules.

Working: the balanced equation is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, so 6 molecules of carbon dioxide and 6 molecules of water make 1 molecule of glucose. 12 ÷ 6 = 2, so 12 molecules of carbon dioxide and 12 molecules of water make 2 molecules of glucose (and 12 molecules of oxygen). Check using carbon atoms: 12 CO_2 molecules contain 12 carbon atoms, and each glucose molecule needs 6, so 12 ÷ 6 = 2.

Question B5.04

Starch can easily be broken down to glucose by the enzyme amylase, which is found in plants. Cellulose is much more difficult to break down to glucose. Suggest how these differences between starch and cellulose relate to their functions in a plant.

Answer: Starch is the plant's energy store. When the plant needs energy (for example at night, or when a seed germinates), it must be able to turn the stored starch back into glucose quickly for respiration, so it is useful that amylase can break starch down easily. Cellulose forms the cell walls, which give each cell strength and support and must last for the whole life of the cell. If cellulose were easy to break down (for example by the plant's own amylase), the cell walls would be weakened or destroyed. Because cellulose is difficult to break down, the walls stay strong and permanent.

Question B5.05

Animals do not make or store starch. What is the substance that animals store, which is made of chains of glucose molecules?

Answer: Glycogen (stored mainly in the liver and muscles).

Question B5.06

Explain why some parts of a plant must have sucrose delivered to them.

Answer: Glucose can only be made by photosynthesis in the parts of a plant that contain chlorophyll, which usually means the leaves. Other parts, such as the roots, flowers, fruits, seeds, storage organs and the inner parts of the stem, have no chlorophyll (or get no light), so they cannot make their own glucose. Their cells still need carbohydrate: for respiration to release energy, to make cellulose for new cell walls as they grow, and to make starch for storage. Plants do not transport glucose — they change it into sucrose, which is carried in the phloem to these parts and changed back into glucose when it arrives.

Question B5.07

Copy and complete this table.

Element	……	magnesium
Mineral salt	nitrate ions	magnesium ions
Why needed	to make ……, and then proteins	……
Deficiency	weak growth, yellow leaves	yellowing of leaves, often especially between the veins

Answer:

Element	nitrogen	magnesium
Mineral salt	nitrate ions	magnesium ions
Why needed	to make amino acids, and then proteins	to make chlorophyll
Deficiency	weak growth, yellow leaves	yellowing of leaves, often especially between the veins

The three missing entries are **nitrogen**, **amino acids** and **to make chlorophyll**.

Question B5.08

Making nectar costs a plant energy because it uses up glucose that the plant has made. Explain why the expense is worthwhile.

Answer: Nectar attracts pollinators — insects, bats or birds — to the flowers. The male gametes of a flower are inside pollen grains and cannot move themselves, so the plant relies on these animals to carry pollen from one flower to another. Animals will only do this if they get a reward, and nectar is that reward. Pollination allows fertilisation to happen, so the plant can reproduce sexually and produce seeds that grow into new plants. Without nectar, far fewer flowers would be pollinated, so the glucose used is a worthwhile price for successful reproduction.

Question B5.09

Figure B5.10 shows a photomicrograph (×400) of a small part of a leaf from a tea plant. Identify structures A–D.

Description of Figure B5.10: label A points to the row of cells forming the top surface of the leaf; label B points to the layer of tall, narrow cells packed side by side just below it; label C points to the region of rounded, loosely arranged cells with spaces between them in the lower part of the leaf; label D points up to a gap in the bottom row of cells, with a large space just above it.

Answer:

A — upper epidermis (covered on the outside by the thin cuticle)

B — palisade mesophyll

C — spongy mesophyll

D — a stoma (the opening between two guard cells in the lower epidermis)

Question B5.10

Explain how a palisade mesophyll cell is adapted for its function.

Answer: The function of a palisade mesophyll cell is photosynthesis. It is adapted in these ways:

- It contains very large numbers of chloroplasts, which contain chlorophyll, so it can absorb as much light energy as possible.
- It is tall and narrow, and the cells are packed tightly side by side, so a large number of cells (and chloroplasts) fit into the top part of the leaf, where the light is brightest.
- It is close to the top of the leaf, just below the transparent cuticle and upper epidermis, so it receives plenty of sunlight.
- It is close to the air spaces of the spongy mesophyll, which supply carbon dioxide by diffusion, and to the veins, which supply water.

Question B5.11

Suggest why the cells in the lower epidermis of a leaf (apart from the guard cells) do not contain chloroplasts.

Answer: Very little light reaches the lower epidermis. Sunlight falls mainly on the upper surface of the leaf, and most of it is absorbed by the chloroplasts in the palisade and spongy mesophyll before it reaches the bottom layer. Chloroplasts in the lower epidermis would therefore carry out very little photosynthesis, so making them would waste the plant's materials and energy. The job of the lower epidermis is to cover and protect the leaf and to contain the stomata for gas exchange, not to photosynthesise.

Practice Exam Questions with Answers

Practice Question 1

Identify the raw materials required for photosynthesis. (1 mark)

A) carbon dioxide and oxygen

B) oxygen and glucose

C) glucose and water

D) water and carbon dioxide

Answer: D — water and carbon dioxide. These are the reactants of photosynthesis. Oxygen and glucose are the products, so options A, B and C are wrong.

Practice Question 2

Identify which ion a plant requires to make amino acids from glucose. (1 mark)

A) calcium

B) iron

C) magnesium

D) nitrate

Answer: D — nitrate. Amino acids contain nitrogen, which glucose does not. Plants absorb nitrogen from the soil as nitrate ions and combine it with glucose to make amino acids. Magnesium ions are needed to make chlorophyll, not amino acids.

Practice Question 3

The diagram shows a three-dimensional drawing of a section through a leaf. (9 marks)

Description of the diagram: a block cut from a leaf, with its layers labelled from top to bottom. A labels the thin layer covering the top surface; B labels the top layer of cells; C labels a layer of tall, narrow cells packed side by side; D labels a deep layer of rounded cells with spaces between them; E labels a bundle of tubes running through the leaf; F labels a space between the rounded cells; G labels a pore in the bottom surface; H labels one of the two cells surrounding that pore; I labels the bottom layer of cells.

a) i) Identify the tissue labelled C. [1]

a) ii) Outline how this tissue is adapted for its function. [2]

b) Give the letters of two cells or tissues that do **not** contain chloroplasts. [2]

c) Describe the function of the cell labelled H. [2]

d) Use the letters on the diagram to describe how water reaches a chloroplast in the leaf. [2]

Answer:

a) i) Palisade mesophyll. [1]

a) ii) Any two of the following [2]:

- The cells contain very large numbers of chloroplasts (containing chlorophyll) to absorb as much light energy as possible.
- The cells are tall, narrow and tightly packed side by side, so many cells fit into the top part of the leaf.
- The layer is close to the upper surface, just below the transparent epidermis, so it receives plenty of sunlight.
- The cells are close to air spaces (supplying carbon dioxide) and veins (supplying water).

b) Any two of: **B** (upper epidermis), **I** (lower epidermis, apart from its guard cells) and **E** (vascular bundle — xylem and phloem). [2] Do not choose A, F or G: they are not cells (the cuticle, an air space and a stoma).

c) H is a guard cell. It changes shape to open and close the stoma, G [1]. This controls the diffusion of carbon dioxide into the leaf and oxygen out of it, and controls how much water vapour is lost from the leaf [1].

d) Water arrives in the leaf in the xylem vessels inside the vascular bundle, **E** [1]. It then leaves the xylem and moves from cell to cell by osmosis through the spongy mesophyll, **D**, to the palisade mesophyll cells, **C**, where it enters the chloroplasts [1].

Practice Question 4

The bar chart shows the mean number of chloroplasts in four different types of cell in a leaf. (7 marks)

Data read from the bar chart (y-axis: mean number of chloroplasts per cell, scale 0 to 100; x-axis: type of cell):

Type of cell	Mean number of chloroplasts per cell
A	0 (no bar)
B	about 15
C	about 71
Palisade mesophyll	about 100 (the bar reaches the top of the scale)

a) Explain why palisade mesophyll cells have so many chloroplasts. [2]

b) State which cell could be a cell in the upper epidermis. Explain your answer. [2]

c) Guard cells contain chloroplasts, but not as many as spongy mesophyll cells. Suggest which cell could be a guard cell. [1]

d) Chloroplasts contain chlorophyll. Describe the function of chlorophyll in photosynthesis. [2]

Answer:

a) Palisade mesophyll cells are the main site of photosynthesis in the leaf [1]. They are near the upper surface, where they receive the most light, so having many chloroplasts (and so lots of chlorophyll) lets them absorb as much light energy as possible and photosynthesise quickly, making more glucose [1].

b) Cell **A** [1]. Upper epidermis cells contain no chloroplasts (A has a mean of 0); they are transparent so that light passes through them to the palisade cells below [1].

c) Cell **B** — it has some chloroplasts (about 15), but far fewer than cell C (about 71), which must be a spongy mesophyll cell [1].

d) Chlorophyll absorbs (captures) energy from sunlight [1] and transfers it into chemical energy in carbohydrates — the energy makes carbon dioxide and water react to form glucose [1].

Practice Question 5

(8 marks)

a) Describe how plants synthesise carbohydrates. [4]

b) Outline **two** subsequent uses of the carbohydrates made in photosynthesis. [4]

Answer:

a) Any four of the following [4]:

- Plants make carbohydrates by **photosynthesis**, mainly in the palisade mesophyll cells of their leaves.
- Carbon dioxide diffuses into the leaf from the air through the stomata; water is absorbed from the soil by the roots and carried to the leaf in the xylem.
- In the chloroplasts, **chlorophyll** absorbs energy from sunlight.
- Chlorophyll transfers this energy to carbon dioxide and water, making them react; the reactions are catalysed by enzymes.
- The products are **glucose** (a carbohydrate) and **oxygen**, which is released.
- Equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

b) Any two uses, each with an explanation [2 + 2]:

- **Respiration** [1] — glucose is broken down to release energy, for example for active transport of mineral ions into root hairs, or for building proteins [1].
- **Converted to starch for storage** [1] — starch is insoluble, so it does not affect osmosis or take part in reactions, and it can be quickly turned back into glucose when needed [1].
- **Converted to sucrose for transport** [1] — sucrose is carried in the phloem to parts that cannot photosynthesise, such as the roots, where it is changed back into glucose [1].
- **Converted to cellulose** [1] — to build cell walls for the new cells made as the plant grows [1].
- **Used to make amino acids and then proteins** [1] — glucose is combined with nitrogen from nitrate ions; the proteins are needed for growth [1].
- **Used to make nectar** [1] — nectar attracts pollinators such as insects, so the flowers are pollinated [1].

Practice Question 6

Two students carried out an experiment to investigate the effect of carbon dioxide concentration on the rate of photosynthesis of an aquatic plant. The photograph shows some of the apparatus that they used. (15 marks)

Description of the photograph: several syringes are clamped upright, side by side, outdoors in bright sunshine, with a long thin transparent tube running down from the nozzle of each syringe.

- The students filled five syringes with five different concentrations of sodium hydrogencarbonate solution. This provides carbon dioxide.
- They placed a piece of aquatic plant in each syringe.
- They attached a long transparent tube to the nozzle of each syringe.
- They pushed the plunger of each syringe in a little way, to force some of the sodium hydrogencarbonate into the tubing.
- They then pulled each plunger out a little way, to introduce some air into the tubing.
- They marked the position of the meniscus on each piece of tubing, as shown in the diagram.

Description of the diagram: a short length of the transparent tubing, with water in the upper part and air in the lower part; the meniscus is the boundary between them, and a mark is drawn on the tubing level with the meniscus.

All five syringes were placed in bright sunshine. As the plants photosynthesised, they released oxygen. The oxygen gas collected above the liquid in the syringe. This pushed the liquid down the tube.

The students recorded the distance moved by the meniscus on each piece of apparatus after 30 minutes.

a) Identify the independent variable and the dependent variable in this investigation. [2]

b) Explain why the five syringes were placed in bright sunshine. [2]

The table shows the results of the experiment.

Percentage concentration of sodium hydrogencarbonate solution	0	0.1	0.2	0.3	0.4
Distance moved by meniscus after 30 minutes / mm	5	8	12	14	17

c) Draw a line graph to display these results. Join the points with ruled straight lines. [5]

d) Explain the results of this investigation. [2]

e) i) Suggest **two** possible sources of error in this investigation. [2]

e) ii) Suggest how the investigation could be modified to reduce these sources of error. [2]

Answer:

a) Independent variable: the concentration of the sodium hydrogencarbonate solution, which determines how much carbon dioxide is available [1]. Dependent variable: the distance moved by the meniscus in 30 minutes, which shows how much oxygen was produced, and so the rate of photosynthesis [1].

b) Light is needed for photosynthesis, because it provides the energy [1]. Bright sunshine gives a high light intensity, so light is not a limiting factor: the rate is limited by the carbon dioxide concentration, and all five syringes receive the same high light intensity, making it a fair test [1].

c) The correct graph shows [5]:

- The concentration on the x-axis, labelled "Concentration of sodium hydrogencarbonate solution / %", and the distance on the y-axis, labelled "Distance moved by meniscus after 30 minutes / mm" [1].
- Even scales that use at least half of the grid, for example 0 to 0.4 in steps of 0.1 on the x-axis and 0 to 18 mm in steps of 2 mm on the y-axis [1].
- The five points plotted accurately as small neat crosses at (0, 5), (0.1, 8), (0.2, 12), (0.3, 14) and (0.4, 17) [2 for all five correct; 1 for three or four correct].
- The points joined with ruled straight lines from cross to cross, starting at the first point and ending at the last point, not extended to the origin [1].

The finished line rises steadily from left to right and does not level off.

d) As the concentration of sodium hydrogencarbonate increased from 0% to 0.4%, the distance moved by the meniscus increased from 5 mm to 17 mm [1]. Sodium hydrogencarbonate provides carbon dioxide, a raw material for photosynthesis, so a higher concentration allowed a faster rate of

photosynthesis (carbon dioxide was the limiting factor). More oxygen was produced in 30 minutes, and this collected in the syringe and pushed the liquid further down the tube [1].

The line does not level off, so carbon dioxide was still limiting even at 0.4%. The meniscus moved 5 mm even at 0%, probably because the water contained some dissolved carbon dioxide (or because the gas warmed and expanded in the sunshine).

e) i) Any two of the following [2]:

- The temperature was not controlled: syringes in bright sunshine may warm up, which makes the gas expand (moving the meniscus) and also changes the rate of photosynthesis.
- The light intensity may not have been the same for every syringe, or constant for the whole 30 minutes (clouds, shadows, syringes shading each other).
- The pieces of aquatic plant may have been different sizes or had different numbers of leaves.
- There was only one syringe for each concentration, so there were no repeats to check the results or calculate a mean.
- Some of the oxygen may have dissolved in the solution or stayed stuck to the plant, so it did not move the meniscus.

e) ii) One matching improvement for each error given in part i) [2], for example:

- Stand the syringes in a thermostatically controlled water-bath, checked with a thermometer, to keep the temperature constant.
- Carry out the experiment indoors, with a lamp at the same fixed distance from every syringe (and a heat shield), instead of in sunshine.
- Use pieces of plant of the same length or mass, cut from the same plant.
- Set up three syringes for each concentration and calculate a mean distance.
- Measure the distance against a ruler marked in millimetres, reading at eye level.

Summary of Key Concepts

Chapter Summary

- **Photosynthesis** is a reaction in which plants make carbohydrates from carbon dioxide and water, using light as a source of energy.
- The word equation for photosynthesis is: carbon dioxide + water → glucose + oxygen
- The balanced equation is: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
- **Chlorophyll** is a green pigment present in the chloroplasts of some plant cells. Chlorophyll transfers energy from light into energy in carbohydrates.
- The carbohydrate **glucose** is made in photosynthesis. It can be used to release energy in **respiration**. Glucose can be used to make **starch** for storage of energy and **sucrose** for transport.
- Glucose can be used to make **amino acids and proteins**, using nitrogen obtained from **nitrate ions**. It can also be used to make **chlorophyll**, using nitrogen and also **magnesium ions**.
- **Cellulose** is used to build cell walls, **sucrose** is used for transport in the phloem, and **nectar** attracts insects for pollination.
- Leaves have a **large surface area**, to capture energy from sunlight and to increase absorption of carbon dioxide from the air. **Stomata** on the lower surface allow carbon dioxide to diffuse into the **air spaces**, and then to the cells in the **palisade mesophyll** layer, where most photosynthesis takes place.
- To **test a leaf for starch**, it is first boiled to break down cell membranes, then heated in alcohol to remove the chlorophyll. Finally, iodine solution is added.
- The starch test shows that leaves, or parts of a leaf, that do not have light, chlorophyll or carbon dioxide are not able to photosynthesise and make starch. **Controls** are needed so that we can

compare the results with them.
- We can use an **aquatic plant** to measure the rate of photosynthesis, by counting the number of bubbles given off in unit time, or by measuring the volume of gas given off in unit time.
- This method can be used to investigate the effect of altering **light intensity, temperature or carbon dioxide concentration** on the rate of photosynthesis.

Command Words Used in This Chapter

- **Identify** — name, select or recognise.
- **Give** — produce an answer from a given source or from recall or memory.
- **Outline** — set out the main points.
- **Describe** — state the points of a topic, or give its characteristics and main features.
- **Explain** — set out purposes or reasons; make the relationships between things clear; say why and/or how, and support with relevant evidence.
- **Suggest** — apply your knowledge and understanding to situations where there is a range of valid responses, in order to make proposals or put forward considerations.

Exam Technique Tips

- For "identify" multiple-choice questions, rule out options that contain a product (glucose or oxygen) when the question asks for raw materials.
- For "outline" and "describe" questions, give one clear point per mark — a 2-mark adaptation question needs two separate adaptations.
- For "explain" questions, always link the feature to the function using "so that" or "because", e.g. "many chloroplasts, so that more light energy is absorbed".
- When a question says "use the letters on the diagram", put the letters into your answer (for example "xylem in E, then by osmosis through D to C").
- When describing results, quote numbers with units from the data, then explain the pattern.
- In planning questions, always state the independent variable, the dependent variable, at least two controlled variables, repeats and a safety point.
- On a graph, read values carefully against the gridlines and say "about" if a value lies between lines.

Self-Evaluation Checklist

I can:

- state the word equation for photosynthesis (B5.01)
- write the balanced equation for photosynthesis (B5.01)
- describe the role of chlorophyll in transferring energy from light to carbohydrates, in photosynthesis (B5.01)
- outline the uses of starch, cellulose, sucrose and nectar, made from the carbohydrates made in photosynthesis (B5.01)
- state the functions of nitrate ions and magnesium ions in a plant (B5.01)
- name the different cell types and layers of a leaf (B5.02)
- explain how the features of a leaf are adaptations for photosynthesis (B5.02)
- describe how to investigate the necessity for chlorophyll, light and carbon dioxide for photosynthesis (B5.03)
- explain what a control is, in an experiment (B5.03)
- describe how to investigate the effect of varying light intensity and temperature on the rate of photosynthesis of an aquatic plant (B5.03)